Shashwat Singh

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Research Interests

Morphology Driven Co-Design: Designing morphologically adaptive robots and jointly optimizing mechanical form, fabrication, and control policies for embodied intelligence.

Sim-to-Real Transfer for Multi-Terrain Locomotion: Training reinforcement learning policies in simulation with domain randomization and deploying them zero-shot on physical hardware, using multi-modal sensor fusion for perception and control across unstructured terrains.

Education

Carnegie Mellon University, Pittsburgh, USA — Aug. 2023 – Present
Ph.D. in Robotics
Advisor: Prof. Zeynep Temel

University at Buffalo, Buffalo, NY, USA — Aug. 2021 – Jul. 2023
M.S. in Robotics
Advisor: Prof. Ryan St. Pierre

University of Allahabad, Prayagraj, India — Aug. 2017 – Jun. 2021
B.Tech. in Electronics and Communication

Research Experience

Research Assistant, Carnegie Mellon University, Pittsburgh — Aug. 2023 – Present

Zoom Lab, advised by Prof. Zeynep Temel

- Designed, fabricated, and controlled transformable wheeled robots with real-time morphology adaptation, integrating motion planning and autonomous navigation in unstructured environments.
- Built perception and control stacks that fuse vision, proprioceptive sensing, force/contact feedback, and actuator telemetry into closed-loop control and online adaptation, optimized to run on embedded compute.
- Developed contact-rich physical coupling mechanisms enabling manipulation-style docking in heterogeneous robot teams, working across two universities with mechanical, electrical, and software collaborators.

Research Assistant, University at Buffalo — Aug. 2021 – Jul. 2023

Robot Form and Function Lab, advised by Prof. Ryan St. Pierre

- Designed and fabricated sub-cubic-inch autonomous millirobots with multi-modal locomotion.
- Developed embedded control software under strict size, weight, and power (SWaP) constraints, including real-time sensing, actuation, and communication on custom PCBs.
- Performed dynamic modeling, trajectory optimization, controller design, and experimental validation on hardware, bridging algorithm analysis with physical implementation.

Publications

1. **S. Singh**, S. Zhang, S. Matonis, Z. Temel. "Terraskipper: A centimeter-scale robot for multi-terrain skipping and crawling." *IEEE Intl. Conf. on Robotics and Automation (ICRA)*, 2026.

2. **S. Singh**, J. Ferlazzo, S. Wang, S. N. Boateng, M. Kramer, M. O. Martinez, F. Z. Temel, L. H. Blumenschein. "Physical coupling for collaboration in heterogeneous robot teams." *IEEE Robotics and Automation Letters (RA-L)*, 2026.

3. T. Han, **S. Singh**, S. Patil, Z. Temel. "From fold to function: Dynamic modeling and simulation-driven design of origami mechanisms." *IEEE RoboSoft*, 2026.

4. **S. Singh**, Z. Si, Z. Temel. "PuffyBot: An untethered shape-morphing robot for multi-environment locomotion." *arXiv:2511.09885*, 2025.

5. S. Yi, **S. Singh**, A. Seo, R. St. Pierre, K. Sycara, Z. Temel. "Reconfigurable robot swarms for terrain traversal with passive coupling mechanisms." *Autonomous Robots*, 2025.

6. **S. Singh**, R. Truax, R. St. Pierre. "Buffalo Byte: A highly mobile and autonomous millirobot platform." *IEEE Robotics and Automation Letters (RA-L)*, 2024.

7. **S. Singh**, Z. Temel, R. St. Pierre. "Multi-modal jumping and crawling in an autonomous, springtail-inspired microrobot." *IEEE ICRA*, 2024.

8. S. N. Boateng, **S. Singh**, et al. "Heterogeneous collaboration: A new approach for search and rescue operations." *IEEE Intl. Symp. on Safety, Security, and Rescue Robotics (SSRR)*, 2024.

9. N. Skowronski, M. M. Pour, **S. Singh**, S. J. Longo, R. St. Pierre. "Design and control of jumping microrobots with torque reversal latches." *Bioinspiration & Biomimetics*, 2024.

Awards

Red Whittaker Field Robotics Fellowship, RI, CMU – Recognized for research achievements in field robotics. 2026

Dean's Graduate Achievement Award, SEAS, University at Buffalo – For excellence in research and technical publications. 2023

Best Poster Award – IROS 2026 ROSE (Robotic One Shot Exploration) Workshop, "Palm-Scale Hybrid Aerial-Ground Robot for Confined and Unstructured Environments."

Best Demo Award – IROS 2026 Miniature Multi-terrain and Multi-modal locomotion Workshop, "BaSiL: A Tendon-Driven Beaded Spine Robot for Shape-Adaptive Locomotion."

Best Poster Award – ICRA 2025 Amphibious Robot Workshop, "PuffyBot: An untethered shape morphing robot for multimodal locomotion."

Best Poster Award – ICRA 2024 Unconventional Robot Workshop, "Multi-modal jumping and crawling in an autonomous, springtail-inspired microrobot."

Teaching Experience

Teaching Assistant, Carnegie Mellon University 2025

- 15-482 Autonomous Agents (Fall 2025): led labs and mentored projects in perception, planning, and learning for autonomous robots.
- 16-235 Fantastic Robots and How to Fold Them (Spring 2025): mentored teams in mechanism design, fabrication, and control of origami-inspired robots.

Technical Skills

Machine Learning & AI: Reinforcement Learning (PPO, SAC), Deep Learning, Machine Learning, Imitation Learning, Sim-to-Real Transfer, Optimal Control, PyTorch, TensorFlow
Robotics: Perception, Computer Vision, Multi-Modal Sensor Fusion, Controls, Dynamic Modeling
Programming & Simulation: Python, C++, ROS/ROS2, NVIDIA Isaac Lab/Isaac Sim, MuJoCo
Embedded & Hardware: NVIDIA Jetson Orin, Raspberry Pi, Arduino/microcontrollers, RS-485/Serial,

CAN, sensor integration, PCB design (KiCAD)
Mechanical Design & Fabrication: SolidWorks, Fusion 360, OnShape, CNC machining, rapid prototyping